

- 2 An investigation was carried out to determine the relative concentration of reducing sugars in a range of foods.

In each case, some food was ground up with distilled water. The resulting liquid was filtered to give an extract.

This extract was then tested for the presence of reducing sugars using a reagent.

- (a) (i) Describe how a test for reducing sugars is carried out.

(2)

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- (ii) Explain how this investigation should be carried out to give a semi-quantitative estimate of the concentration of the reducing sugars by comparing the colour of the precipitate with a colour chart.

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(iii) It was assumed that the extract contained all the reducing sugars from the food sample.

Suggest how this assumption could be tested.

(2)

(b) The table shows the results of the investigation.

Food	Colour of precipitate
apple	brick red
bread	green
cashew nut	no precipitate
potato	yellow
sweet potato	orange

Complete the table below by writing the names of the foods in order from the highest concentration of reducing sugars to the lowest concentration.

(2)

highest —————> lowest

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- (c) The reagent used to test for reducing sugars is a blue solution containing soluble copper ions. Reducing sugars react with the copper to form an insoluble precipitate.

After testing, the precipitate formed can be removed. This leaves a solution that is less blue.

- (i) Explain which food extract would leave the least blue solution when the precipitate is removed.

(3)

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